

Figure 9.2 Fixed-lag smoothing. We desire an estimate of the state at each time step based on measurements two time steps ahead. After processing y_2 , we form the estimate $\hat{x}_{0,2}$, which is the estimate of x_0 based on measurements up to and including y_2 . Similarly, $\hat{x}_{1,3}$ is the estimate of x_1 based on measurements up to and including y_3 .

some data that are available for post-processing. For example, if a manufacturing process has run over the weekend and we have recorded all of the data, and now we want to plot a time history of the best estimate of the process state, we can use all of the recorded data to estimate the states at each of the time points. Fixed-interval smoothing is depicted in Figure 9.3 and is discussed in Section 9.4.

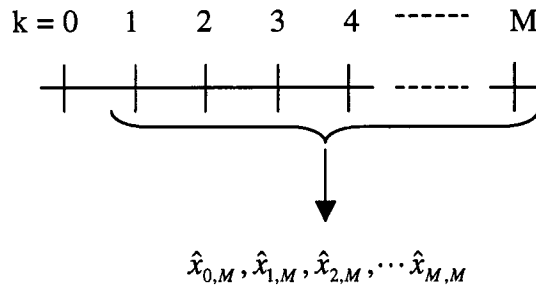


Figure 9.3 Fixed-interval smoothing. We desire an estimate of the state at each time step based on all of the measurements in some interval. After processing all of the measurements from y_1 to y_M , we form the estimate $\hat{x}_{0,M}$, which is the estimate of x_0 based on all the measurements. Similarly, $\hat{x}_{1,M}$ is the estimate of x_1 based on all the measurements.

Our derivation of these optimal smoothers will be based on a form for the Kalman filter different than we have seen in previous chapters. Therefore, before we can discuss the optimal smoothers, we will first present an alternate Kalman filter form in Section 9.1.

9.1 AN ALTERNATE FORM FOR THE KALMAN FILTER

In order to put ourselves in position to derive optimal smoothers, we first need to derive yet another form for the Kalman filter. This is the form presented in [And79]. The equations describing the system and the Kalman filter were derived in Section 5.1 as follows:

$$\begin{aligned}
x_k &= F_{k-1}x_{k-1} + w_{k-1} \\
y_k &= H_k x_k + v_k \\
P_{k+1}^- &= F_k P_k^+ F_k^T + Q_k \\
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \\
P_k^+ &= (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k R_k K_k^T \\
\hat{x}_k^- &= F_{k-1} \hat{x}_{k-1}^+ \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - H_k \hat{x}_k^-)
\end{aligned} \tag{9.3}$$

Now if we define L_k as

$$\begin{aligned}
L_k &= F_k K_k \\
&= F_k P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1}
\end{aligned} \tag{9.4}$$

and substitute the expression for $\hat{x}_k^+$ into the expression for $\hat{x}_{k+1}^-$, then we obtain

$$\begin{aligned}
\hat{x}_{k+1}^- &= F_k \hat{x}_k^- + F_k K_k (y_k - H_k \hat{x}_k^-) \\
&= F_k \hat{x}_k^- + L_k (y_k - H_k \hat{x}_k^-)
\end{aligned} \tag{9.5}$$

Expanding the expression for P_k^+ gives

$$P_k^+ = P_k^- - K_k H_k P_k^- - P_k^- H_k^T K_k^T + K_k H_k P_k^- H_k^T K_k^T + K_k R_k K_k^T \tag{9.6}$$

Substituting for K_k gives

$$\begin{aligned}
P_k^+ &= P_k^- - P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^- - \\
&\quad P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^- + \\
&\quad P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^- + \\
&\quad P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} R_k (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^-
\end{aligned} \tag{9.7}$$

Performing some factoring and collection of like terms on this equation gives

$$\begin{aligned}
P_k^+ &= P_k^- + P_k^- H_k^T [-(H_k P_k^- H_k^T + R_k)^{-1} - (H_k P_k^- H_k^T + R_k)^{-1} + \\
&\quad (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} + \\
&\quad (H_k P_k^- H_k^T + R_k)^{-1} R_k (H_k P_k^- H_k^T + R_k)^{-1}] H_k P_k^- \\
&= P_k^- + P_k^- H_k^T [-(H_k P_k^- H_k^T + R_k)^{-1} - (H_k P_k^- H_k^T + R_k)^{-1} + \\
&\quad (H_k P_k^- H_k^T + R_k)^{-1} (H_k P_k^- H_k^T + R_k) (H_k P_k^- H_k^T + R_k)^{-1}] H_k P_k^- \\
&= P_k^- - P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^-
\end{aligned} \tag{9.8}$$

Substituting this expression for P_k^+ into the expression for P_{k+1}^- gives

$$\begin{aligned}
P_{k+1}^- &= F_k P_k^+ F_k^T + Q_k \\
&= F_k [P_k^- - P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} H_k P_k^-] F_k^T + Q_k \\
&= F_k P_k^- (F_k - L_k H_k)^T + Q_k
\end{aligned} \tag{9.9}$$